The Trivial and Sign Weighted Bergman Modules Are Never Similar

Run `fa_banach_001` – lane 1

17 August 2026

Status: candidate full negative solution. For every $n \geq 2$ and every $\lambda \geq 1$, the trivial and sign reducing submodules of $\mathbb{A}^{(\lambda)}(\mathbb{D}^n)$ are not similar over $\mathbb{C}[z_1, \dots, z_n]^{\mathfrak{S}_n}$. Multiplication by the Vandermonde is also never onto. This settles the general questions posed in Section 4.3 and Remark 4.22 of arXiv:2409.11101.

1 Source question and scope

Biswas, Ghosh, Narayanan, and Shyam Roy ask in Section 4.3 of *Contractive Hilbert modules on quotient domains*, arXiv:2409.11101 [1], whether the trivial and sign submodules $\mathbb{A}^{(\lambda)}_{\text{triv}}(\mathbb{D}^n)$ and $\mathbb{A}^{(\lambda)}_{\text{sign}}(\mathbb{D}^n)$ are similar over the symmetric polynomial ring. They prove non-similarity only for $n = 2$ and $\lambda = 1, 2$.

> The last equality holds only if $\lambda = 1$, which leads us to a contradiction as $\lambda > 1$. □
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> 4.3. **Similarity.** In view of Proposition 4.19, it is natural to ask whether $\mathbb{A}^{(\lambda)}_{\text{triv}}(\mathbb{D}^n)$ and $\mathbb{A}^{(\lambda)}_{\text{sign}}(\mathbb{D}^n)$ are similar over $\mathbb{C}[z_1, \dots, z_n]^{\mathfrak{S}_n}$. We show for $\lambda = 1, 2$ the submodules $\mathbb{A}^{(\lambda)}_{\text{triv}}(\mathbb{D}^2)$ and $\mathbb{A}^{(\lambda)}_{\text{sign}}(\mathbb{D}^2)$ are not similar over $\mathbb{C}[z_1, z_2]^{\mathfrak{S}_2}$. We start with the following lemma.

On PDF page 21, Remark 4.22 asks the corresponding general division question for the Vandermonde V_n and again records only the case $n = 2$, $\lambda = 1, 2$.

> To prove the assertion for $\lambda = 1$, we simply note from Equation (4.8) that Equation (4.9) reduces to $m^2 \leq 4K^2$ for all $m \geq 1$. This contradiction completes the proof. □
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> **Remark 4.22.** In general, the map $f \mapsto V_n f$, where $V_n(z) = \prod_{i<j}(z_i - z_j)$, from $\mathbb{A}^{(\lambda)}_{triv}(\mathbb{D}^n)$ to $\mathbb{A}^{(\lambda)}_{sign}(\mathbb{D}^n)$ is one-one and one ask whether this map is onto or not to have a similarity. It turns out that this is the same is asking whether V_n is a 'divisor' in the sense of [7, 25] - that is, if $f \in \mathbb{A}^{(\lambda)}(\mathbb{D}^n)$ and $f/V_n \in \mathcal{O}(\mathbb{D}^n)$, where $\mathcal{O}(\mathbb{D}^n)$ is the algebra of holomorphic functions on $\mathbb{D}^n$, then is it true that $f/V_n \in \mathbb{A}^{(\lambda)}(\mathbb{D}^n)$? The Proposition above shows that the division problem fails for $\lambda = 1, 2$ when $n = 2$.

We answer both questions for all $n \geq 2$ and $\lambda \geq 1$. In fact, the kernel argument below works for every $\lambda > 0$ for which the RKHS with the displayed kernel is considered.

2 Definitions and proof intuition

Let $\mathcal{H} = \mathbb{A}^{(\lambda)}(\mathbb{D}^n)$ be the reproducing-kernel Hilbert space with kernel

$$K_\lambda(z, w) = \prod_{j=1}^{n} (1 - z_j \overline{w_j})^{-\lambda}.$$

Let $\mathcal{H}_+$ and $\mathcal{H}_-$ denote its symmetric and anti-symmetric subspaces. They are the trivial and sign reducing submodules for the action of $\mathfrak{S}_n$. Put

$$\Delta(z) = \prod_{1 \le i < j \le n} (z_i - z_j), \qquad N = \frac{n(n-1)}{2}.$$

Every anti-symmetric holomorphic function is divisible by Δ, with a symmetric holomorphic quotient. We therefore introduce

$$\widetilde{\mathcal{H}}_- = \{g \text{ symmetric and holomorphic} : \Delta g \in \mathcal{H}_-\}, \qquad \|g\|_{\widetilde{\mathcal{H}}_-} = \|\Delta g\|_{\mathcal{H}}.$$

Multiplication by Δ is a unitary from $\widetilde{\mathcal{H}}_-$ onto $\mathcal{H}_-$ and respects multiplication by symmetric polynomials.

Proof intuition

The two divided modules have dramatically different point-evaluation sizes near the boundary of the full diagonal. At $\boldsymbol{d}(a) = (a, \ldots, a)$, the kernel norm in $\widetilde{\mathcal{H}}_-$ exceeds the one in $\mathcal{H}_+$ by the exact factor $(1 - |a|^2)^{-N}$, up to a fixed constant. A module similarity must be multiplication by a holomorphic function ψ, and its inverse must be multiplication by $h = 1/\psi$. Kernel-vector estimates then force $|h(\boldsymbol{d}(a))| \lesssim (1 - |a|^2)^N$ uniformly in the argument of a. Maximum modulus says that the holomorphic function $a \mapsto h(a, \ldots, a)$ must be zero, contradicting $h\psi = 1$.

3 The diagonal kernel calculation

Let L_+ and L_- be the reproducing kernels of $\mathcal{H}_+$ and $\widetilde{\mathcal{H}}_-$, respectively.

Lemma 1. *For $a \in \mathbb{D}$,*

$$L_+(\boldsymbol{d}(a), \boldsymbol{d}(a)) = (1 - |a|^2)^{-n\lambda},$$
$$L_-(\boldsymbol{d}(a), \boldsymbol{d}(a)) = c_{n,\lambda}(1 - |a|^2)^{-n\lambda - 2N},$$

where

$$c_{n,\lambda} = \frac{1}{n!} \prod_{j=0}^{n-1} \frac{(\lambda)_j}{j!} > 0.$$

Consequently,

$$\left(\frac{L_-(\boldsymbol{d}(a), \boldsymbol{d}(a))}{L_+(\boldsymbol{d}(a), \boldsymbol{d}(a))} \right)^{1/2} = \sqrt{c_{n,\lambda}}(1 - |a|^2)^{-N}.$$

Proof. The symmetric and anti-symmetric projection formulas give

$$L_+(z, w) = \frac{1}{n!} \operatorname{per} \left[(1 - z_i \overline{w_j})^{-\lambda} \right]_{i,j=1}^{n}, \tag{1}$$

$$L_-(z, w) = \frac{1}{n!} \frac{\det \left[(1 - z_i \overline{w_j})^{-\lambda} \right]_{i,j=1}^{n}}{\Delta(z)\overline{\Delta(w)}}. \tag{2}$$

The quotient in (2) extends holomorphically across the collision sets because the numerator is alternating separately in z and $\overline{w}$. At $w = \boldsymbol{d}(a)$, every summand of (1) agrees, which gives the first formula.

For the second formula, first coalesce all variables at zero. The power series

$$(1 - zu)^{-\lambda} = \sum_{k\geq 0} \frac{(\lambda)_k}{k!} z^k u^k$$

and the Cauchy–Binet expansion show that the lowest nonzero alternating bidegree of the determinant uses $k = 0, 1, \ldots, n-1$. Dividing by the two Vandermondes and letting all variables tend to zero yields

$$\prod_{j=0}^{n-1} \frac{(\lambda)_j}{j!}.$$

For a general a, apply the disc automorphism $\phi_a(z) = (z - a)/(1 - \overline{a}z)$ to every coordinate. The identities

$$1 - \phi_a(z)\overline{\phi_a(w)} = \frac{(1 - |a|^2)(1 - z\overline{w})}{(1 - \overline{a}z)(1 - a\overline{w})}$$

and

$$\phi_a(z_i) - \phi_a(z_j) = \frac{(1 - |a|^2)(z_i - z_j)}{(1 - \overline{a}z_i)(1 - \overline{a}z_j)}$$

transform the coalesced determinant quotient at zero into $(1 - |a|^2)^{-n\lambda - 2N}$ times the same constant. The factor $1/n!$ in (2) completes the proof. □

4 No module similarity

Theorem 2. *For every $n \geq 2$ and $\lambda \geq 1$, the Hilbert modules $\mathbb{A}^{(\lambda)}_{\mathrm{triv}}(\mathbb{D}^n)$ and $\mathbb{A}^{(\lambda)}_{\mathrm{sign}}(\mathbb{D}^n)$ are not similar over $\mathbb{C}[z_1, \ldots, z_n]^{\mathfrak{S}_n}$.*

Proof. Via division by Δ, it suffices to rule out a bounded invertible module map $X : \mathcal{H}_+ \to \widetilde{\mathcal{H}}_-$. Symmetric polynomials are dense in $\mathcal{H}_+$, so 1 is cyclic. They are also dense in $\widetilde{\mathcal{H}}_-$: anti-symmetric polynomials are dense in $\mathcal{H}_-$, and every such polynomial is Δ times a symmetric polynomial.

Set $\psi = X1$. For every symmetric polynomial p, the module relation gives $Xp = p\psi$. Density and continuity of point evaluations extend this identity to every $f \in \mathcal{H}_+$, so $X = M_\psi$. Applying the same argument to $Y = X^{-1}$ gives $Y = M_h$ for a symmetric holomorphic h. Hence

$$h\psi = 1.$$

Let k_w^+ and k_w^- be the kernel vectors of $\mathcal{H}_+$ and $\widetilde{\mathcal{H}}_-$. Adjointing the two multiplier identities gives

$$M_\psi^* k_w^- = \overline{\psi(w)} k_w^+, \qquad M_h^* k_w^+ = \overline{h(w)} k_w^-.$$

Therefore, with $q(w) = \sqrt{L_-(w,w)/L_+(w,w)}$,

$$\frac{q(w)}{\|Y\|} \leq |\psi(w)| \leq \|X\| q(w).$$

Put $w = \boldsymbol{d}(a)$ and use Lemma 1. Since $h = 1/\psi$,

$$|h(a, \ldots, a)| \leq \frac{\|Y\|}{\sqrt{c_{n,\lambda}}} (1 - |a|^2)^N.$$

The function $g(a) = h(a, \ldots, a)$ is holomorphic on $\mathbb{D}$. For every $0 < r < 1$, the preceding estimate holds uniformly on $|a| = r$, and its right-hand side tends to zero as $r \uparrow 1$. By the maximum-modulus principle, g vanishes identically. This contradicts $h\psi = 1$ on the full diagonal. □

Corollary 3. *For every $n \geq 2$ and $\lambda \geq 1$, multiplication by Δ from $\mathcal{H}_+$ to $\mathcal{H}_-$ is not onto. Equivalently, there exists $f \in \mathbb{A}^{(\lambda)}(\mathbb{D}^n)$ such that f/Δ is holomorphic on $\mathbb{D}^n$ but $f/\Delta \notin \mathbb{A}^{(\lambda)}(\mathbb{D}^n)$.*

Proof. In the divided model, multiplication by Δ becomes the identity map $J : \mathcal{H}_+ \to \widetilde{\mathcal{H}}_-$. It is bounded because Δ is a polynomial multiplier. If it were onto, the open mapping theorem would make J^{-1} bounded. Applying $(J^{-1})^*$ to kernel vectors would force $L_-(w,w) \leq \|J^{-1}\|^2 L_+(w,w)$, contrary to Lemma 1 as $w = \boldsymbol{d}(a)$ and $|a| \uparrow 1$. Finally, every anti-symmetric holomorphic function is holomorphically divisible by Δ, so failure of surjectivity has exactly the stated division interpretation. $\square$

5 Verification, limitations, and novelty

The accompanying script forms the mixed-derivative confluent matrix of $(1 - zu)^{-\lambda}$ and verifies the determinant formula in Lemma 1 for eleven exact cases with $2 \leq n \leq 5$ and $\lambda \in \{1, 2, 5/2\}$. This audits the constant and exponent; it is not used as a proof.

The conclusion concerns the trivial and sign representations of $\mathfrak{S}_n$ and the full parameter range in the source question. It does not classify similarity among higher-dimensional irreducible-representation summands or among modules for other reflection groups.

A bounded novelty search on 17 August 2026 used arXiv, the local full-source corpus, exact title/author searches, the phrases "trivial sign weighted Bergman similarity", "Vandermonde division polydisc", and citation searches for arXiv:2409.11101. It found the source paper and earlier unitary inequivalence work, but no all-parameter similarity or division answer. Of particular significance, Sarkar's December 2025 survey [2] still presents Vandermonde surjectivity/division on $\mathbb{D}^n$ as an open problem. Thus the packet should receive expert review as a plausible new full result.

Human-review recommendation

High priority. Check (i) the normalization and covariance in Lemma 1, and (ii) the claim that every bounded module isomorphism is a multiplier after the Vandermonde division. These are the only substantive hinges; the final maximum-modulus contradiction is then immediate.

References

[1] S. Biswas, G. Ghosh, E. K. Narayanan, and S. Shyam Roy, *Contractive Hilbert modules on quotient domains*, arXiv:2409.11101 (2024).

[2] J. Sarkar, *A compendium of research in operator algebras and operator theory*, arXiv:2512.20979 (2025), Section 1, "On similarity."